

programming language. Scilab is a numerical computational package and numerical programming language. Scratch is a visual programming language for teaching children programming. SQL (Structured Query Language) is a programming language used for relational database management. Simula and Smalltalk may not be relevant today but they have influenced the development of programming languages like C++, Java, etc.

T for TypeScript

TypeScript is an open source programming language released under Apache License 2.0 and is developed by Microsoft. It has the syntax and all the features of JavaScript, with some additional features like static typing. TypeScript has been gaining popularity in recent years. In the *Open Source For You* article mentioned earlier, TypeScript has been included as a promising programming language. Tcl (Tool Command Language), an interpreted dynamic programming language is also worth a mention in this category.

U for Umple

This letter too doesn't have much to offer. But for the sake of completion, Umple represents the letter U in our alphabet. Umple is an object-oriented programming language used for modelling with class diagrams and state diagrams. It was first released in 2008, with the latest stable release in 2018. But do remember that the UNIX shell being Turing complete qualifies as a programming language.

V for Verilog

Consider any decent programming language that has some followers and prefix the term with the word 'visual' and you will get a new programming language! I am not joking – we have Visual Basic, Visual Basic.NET, Visual C++, Visual FoxPro, and many

more as proof. In spite of this vast choice, I prefer VHDL (Very High Speed Integrated Circuit Hardware Description Language) or Verilog, both hardware description languages. My description of hardware, often limited to the price, made the choice very difficult. A cursory glance over a few articles made me believe that Verilog leads the race between the two marginally and hence the choice. But I will be very happy to switch sides if somebody could convince me why VHDL is better than Verilog.

W for Wolfram Language

Wolfram language is yet another proprietary commercial programming language in our list. But I believe I am justified in my choice because the other programming languages in this category have hardly made any impact in the programming world. Wolfram Language is a multi-paradigm programming language developed by Wolfram Research, and is the programming language of the mathematical symbolic computation program Mathematica.

X for XOTcl

My ns-2 (network simulator-2) simulation days made me choose XOTcl from an otherwise insignificant list of programming languages. XOTcl is an object-oriented extension for the Tool Command Language (Tcl). XOTcl is used for topology generation in ns-2.

Y for Yorick

Y is the least promising letter in our programming languages alphabet. Yoix, a dynamic programming language and Yorick, an interpreted programming language designed for numeric and

scientific computing, are the only contenders in this category. I have chosen Yorick over Yoix because its latest stable release was four years ago, compared to seven years for Yorick.

Z for Zsh

Z is another letter with no significant programming languages in the list. But since any tool that is Turing complete qualifies as a programming language, I am nominating Z Shell (Zsh). Z shell is an interactive login shell and a command interpreter for UNIX shell scripting. In the latest major release for Apple Macintosh computers, macOS Catalina has Zsh as the shell instead of Bash.

The selection of a particular programming language is based a lot on personal preferences of the author but efforts have been made to mention all the significant programming languages. To my knowledge there are programming languages named B, C, D, E, F, J, K, P, Q, R, S and T but many of them are old or obscure with no practical use. I will be happy if somebody could point out the programming languages I may have missed in this list. Careful analysis tells us that C, J, M, P, R and S are the most important letters in our programming languages alphabet. As a fun fact, there are programming languages named after animals (Pig, Python and Cobra), places (Java), famous people (Ada, Babbage, Bertrand, Orwell and Pascal), things (Boomerang, Dart and Ruby), etc. But apart from all this fun, our programming languages alphabet clearly tells us that the options are unlimited when it comes to building applications and a career. **END** 

By: Deepu Benson

The author is a free software enthusiast whose area of interest is theoretical computer science. The open source tools of his choice include ns-2 and ns-3. The author maintains a technical blog at www.computingforbeginners.blogspot.in.